Response to reviewers

Introspection Dynamics with Mutation in Additive Games

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Dear Professor Zaccour,

We thank both reviewers for their positive assessments and for a further set of careful comments, all of which we have addressed. We respond to each comment below; reviewer text is set in italics, and our responses follow in upright type.

Reviewer 1

> *Page 3: "At each step, one player i is selected uniformly at random, and draws an alternative action b uniformly from the $L_i - 1$ actions other than a_i." This sentence can be confusing because it sounds as if a player cannot mutate to the current action. And, to fully describe the process, you should also define the transition probability for when $b = a_i$.*

Done, on both counts. The process is now described as an explicit two-stage update: with probability μ_{ib} the selected player mutates to action b, and the target may be their current action, in which case the state does not change; otherwise, with probability $1 - M_i$, they introspect, drawing an alternative uniformly from the other $L_i - 1$ actions and switching with probability given by the Fermi function. The holding probability $T^{\text{intro}}_{\mathbf{a},\mathbf{a}}$ is now displayed as a numbered equation immediately after the transition probability, with a sentence stating which events it collects (mutation to the current action, and introspection without switching).

> *Page 4: the m_i notation for the stationary distribution could be understood as a scalar, when it's a vector. Consider using boldface.*

Done. The stationary distribution is now written $\mathbf{m}_i$ throughout, consistent with the boldface state $\mathbf{a}$.

> *Page 4: in the last equation (after equation 6), $T_{\mathbf{a},\mathbf{a}}$ doesn't seem obvious to me (at least, I had to verify it on the side). Or perhaps I missed something.*

The reviewer did not miss anything; the expression was stated without justification. It now follows by inspection: the holding probability is the numbered equation added in response to the first comment, and the proof of Theorem 1 states that the expression follows from it together with the diagonal entries $K_i(a_i, a_i) = 1 - \sum_{b \neq a_i} K_i(a_i, b)$.

> *Page 5: in Theorem 2's proof, you refer to equation 5 when I think you meant the equation next to 5 (which is not numbered).*

Quite right: the intended reference was the formula for the off-diagonal entries of the selection kernel K_i, which does not carry the $1/N$ factor. That formula is now a numbered equation and the proof cites it.

> *Page 5: "By Theorem 1 the stationary marginal of player i's action is m_i, so the long-run probability that player i cooperates is $m_i(C) = p_i$." I think Theorem 1 doesn't even need to be invoked here, right?*

Right again. The proof now gives the direct argument: additivity decouples player i from the rest of the population, so their action evolves as a two-state Markov chain in its own right, with kernel $(1-\frac{1}{N})I+\frac{1}{N}K_i$, which has the same stationary distribution as K_i. Theorem 1 is no longer invoked.

> *Page 5: "Additivity makes a player's choice forgetful:..." maybe make it clear here again that is both additivity AND having only 2 actions that makes a player's choice 'forgetful'.*

Done, and the comment prompted us to correct a related error. The sentence now reads "Additivity and the restriction to two actions together make a player's choice forgetful". In addition, the property is now named and stated where it is introduced, rather than only being pointed at ahead of Theorem 2: it is the property that the two rows of K_i coincide, so that the probability that a selected player's next action is C does not depend on the action they currently hold. We had justified its failure for $L_i \geq 3$ by saying that the probability of retaining the current action depends on that action, which is mistaken, since this is equally true of the two-action case, where the retention probabilities are p_i and $1-p_i$. The text now gives the correct reason in both directions: with a single alternative the two switching probabilities are complementary, by $\phi_i(x)+\phi_i(-x) = 1$, and it is this that aligns the rows; for $L_i \geq 3$ the rows always differ, since identical rows would force g_i to be constant, and equating a diagonal entry with the off-diagonal entries in its column would then require $L_i = 2$. The closed form of Theorem 2 is thus special to two actions in a strict sense, and not merely generically.

> *Page 16: I assume you use some standard correlation measure (Pearson for binary variables?) but it would be nice to specify which one is exactly used.*

Done. Both the body text and the caption of the figure now state that $\mathrm{corr}(a_i, a_j)$ is the Pearson correlation coefficient of the binary actions under the exact stationary distribution (for indicator variables, the phi coefficient), which is what the accompanying source code computes.

Reviewer 2

> *First about Equation (15). I believe Eq. (15) corresponds to the sentence "[...] each unit contributed changes the marginal value of the next by a factor $w > 0$ [...]", only when the object $X(\mathbf{a})$ is an integer (i.e., the contributions α_i are integers). For continuous values of contributions, there has to be some other normalization that is not $(w-1)$. Therefore, if you want to stick to continuous values of alpha, I suggest that you mention that this formula is just a canonical extension of the 'synergy' model. Or, perhaps you would like to derive the other formula from first principle or mention that alpha is restricted to integers only.*

The reviewer is correct, and we have adopted their first suggestion. The text now states that the geometric-sum reading requires the total contribution $X(\mathbf{a})$ to be an integer, as it is in both of our examples ($\alpha_i = 1$ for the top row of the figure and $\alpha_i = i$ for the correlation panels), and that for non-integer contributions we regard the payoff formula as the canonical analytic extension of the synergy and discounting model, with the normalisation $w - 1$ retained so that the linear game is still recovered as $w \to 1$.

> *Finally, I was wondering what the purpose of studying this model is, broadly speaking (in evolutionary terms). For example, in standard games (let's just say in additive games or the non-linear public goods game that the authors study) would players ever learn to be more exploratory (i.e., would this mutational model itself evolve and spread in a population that does only pure introspection selection)? Does incorporating mutations in decisions have a fitness advantage? Can this be the case or is this model simply dominated as a strategy if the co-player is not exploring? I would have liked if the authors commented on that aspect. It is also quite easy to perform analysis on this since the authors already have payoff formulas.*

We thank the reviewer for this question, and we agree that the closed forms make the analysis tractable. Indeed, the additive structure gives a sharp preliminary answer. By the additive decomposition and the product measure, a player's expected stationary payoff is affine in their own cooperation probability p_i, with slope $-\delta_i$, whilst the co-players' marginals are entirely beyond their influence. In a social dilemma ($\delta_i > 0$) symmetric exploration raises p_i above the pure-introspection value $\phi_i(\delta_i) < \frac{1}{2}$, so it strictly lowers the explorer's own payoff whatever the co-players do: unbiased exploration is dominated, confirming the reviewer's conjecture, and in a form stronger than conjectured, since by the decoupling it does not matter whether the co-players explore. Two further observations follow from the same formulas. First, what is dominated is exploration towards the disfavoured action rather than noise as such: mutation biased towards defection pushes p_i below $\phi_i(\delta_i)$ and outperforms pure introspection. Second, exploration towards cooperation benefits every co-player, so it is altruistic, and its evolution inherits the structure of the underlying dilemma. Beyond additivity the decoupling fails, and our Figure 4 shows that cooperation can peak at a strictly positive mutation rate, so there exploration may plausibly carry an individual advantage.

We believe a proper treatment of this question, with heritable update rules competing on the slower evolutionary timescale, and the non-additive case examined, is a study in its own right, and, mindful that the manuscript already grew substantially in the previous round, we have not added it to the paper. Instead the conclusion now closes with a new paragraph that does two things. It first makes the evolutionary interpretation of our results explicit: the mutation rates model trembles, experimentation, and behavioural noise, and on this reading the results describe how noisy self-reflection shapes long-run behaviour, keeping every action in play, confining each player strictly inside the pure conventions however strong selection becomes, and leaving cooperation at a mutation-selection balance. It then poses the reviewer's question, whether such noise would itself be selected for if update rules were heritable and fitness were the expected stationary payoff, citing the exploration dynamics of Traulsen et al. (2009), and notes that the closed forms make it tractable and that it is a natural direction for future work.

Thank you again for the constructive reports.

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